PRACTICE SHEET

The decimal equivalent of $(101011)_2$ is:
(a) $(43)_{10}$ (b) $(59)_{10}$
(c) $(47)_{10}$ (d) None of these

1. 100101 + 101 + 1101 + 100 is equal to:
(a) 1101001 (b) 100000
(c) 111011 (d) None of these

2. What is the decimal number representation of the binary number $(11101.001)_2$?
(a) 30.125 (b) 29.025
(c) 29.125 (d) 28.025

3. Binary fraction 0.1011 in decimal system is:
(a) 0.6875 (b) 0.8675
(c) 0.7685 (d) None of these

4. The decimal number 0.77 in binary system is:
(a) $(0.1100101)_2$ (b) $(0.11000101)_2$
(c) $(0.10100101)_2$ (d) None of these

5. The binary number 10110100001 in decimal system is:
(a) 441 (b) 1441
(c) 1241 (d) 241

6. The complement of $(1011100)_2$ is:
(a) $(1010100)_2$ (b) $(0111100)_2$
(c) $(0100011)_2$ (d) $(1000011)_2$

7. 2's complement of $(1011001100)_2$ is:
(a) $(0100110100)_2$ (b) $(0100110010)_2$
(c) $(0100110101)_2$ (d) $(0100110111)_2$

8. What is the decimal equivalent of $(101.101)_2$?
(a) $(5.225)_{10}$ (b) $(5.525)_{10}$
(c) $(5.625)_{10}$ (d) $(5.65)_{10}$

9. If the sum of the binary numbers $(11011)_2$, $(10110110)_2$ and $(10011x0y)_2$ is the binary number $(101101101)_2$, then the values of x and y respectively, are:
(a) 1 and 1 (b) 1 and 0
(c) 0 and 1 (d) 0 and 0

10. What is the value of X, if $(1010)_2 \times (111)_2 = (X)_{10}$?
(a) 60 (b) 70
(c) 75 (d) 80

11. What is $(1111)_2 + (1001)_2 - (1010)_2$ equal to?
(a) $(111)_2$ (b) $(1100)_2$
(c) (1110)2 (d) (1010)2

12. The difference of two numbers $(1100110011)_2$ and $(1101001011)_2$ in binary system is:
(a) $(100000)_2$ (b) $(101010)_2$
(c) (11000)2 (d) (10111)2

13. The maximum three digit integer in the decimal system will be represented in the binary system by which one of the following?
(a) $(1111110001)_2$ (b) $(111111110)_2$
(c) (1111100111)2 (d) (1111000111)2

14. What is $(1111)_2 + (1001)_2 - (1010)_2$ equal to?
(a) $(111)_2$ (b) $(1100)_2$
(c) (1110)2 (d) (1010)2

15. What is the binary number equivalent of the decimal number 32.25?
(a) 100010.10 (b) 100000.10
(c) 100010.01 (d) 100000.01

16. What is the equivalent binary number of the decimal number 13.625?
(a) 1101.111 (b) 1111.101
(c) 1101.101 (d) 1111.111

17. The number 0.0011 in binary system represents?
(a)Rational number 3/8 in decimal system
(b)Rational number 1/8 in decimal system
(c)Rational number 3/16 in decimal system
(d)Rational number 5/16 in decimal system

18. When is the value of:

$$\frac{(0.101)_2^{(11)_2} + (0.011)_2^{(11)_2}}{(0.101)_2^{(10)_2} - (0.101)_2^{(01)_2}(0.011)_2^{(01)_2} + (0.011)_2^{(10)_2}}$$

(a) $(0.001)_2$ (b) $(0.01)_2$
(c) (0.1)2 (d) (1)2

19. If $x=(1101)_2$ and $y = (110)_2$, then what is the value of $x^2 - y^2$?
(a) $(1000101)_2$ (b) $(10000101)_2$
(c) $(10001101)_2$ (d) $(10010101)_2$

ANSWER KEYS																			
1.	a	**2.**	c	**3.**	c	**4.**	a	**5.**	b	**6.**	b	**7.**	c	**8.**	a	**9.**	c	**10.**	b
11.	b	**12.**	c	**13.**	c	**14.**	c	**15.**	c	**16.**	d	**17.**	c	**18.**	c	**19.**	d	**20.**	b

Solutions

Sol. 1. **(a)**
$(101011)_2 = (?)_{10}$
$=[1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0]_{10}$
$=[32+0+8+0+2+1]_{10} = (43)_{10}$

Sol. 2. **(c)**
1 0 0 1 0 1
+ 1 0 1
+ 1 1 0 1
+ 1 0 0

= 1 1 1 0 0 1 1

Sol. 3. **(c)**
$(11101.001)_2 = (?)_{10}$
$=[1 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 + 0 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3}]_{10}$
$=[16 + 8 + 4 + 1 + 0.125]_{10} = (29.125)_{10}$

Sol. 4. **(a)**
$=[1 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3} + 1 \times 2^{-4}]_{10}$
=11/16=0.6875

Sol. 5. **(b)**

2	0.77	
2	1.54	1
2	1.08	1
2	0.16	0
2	0.32	0
2	0.64	0
	1.28	1
	0.56	0
	1.12	1

$(0.77)_2 = (0.11000101)_{10}$

Sol. 6. **(b)**
$[1 \times 2^{10} + 0 \times 2^9 + 1 \times 2^8 + 1 \times 2^7 + 0 \times 2^6 + 1 \times 2^5 + 0 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0]_{10}$
=[1024 + 256 + 128 + 32 + 1] = 1441

Sol. 7. **(c)**
complement of 1011100 is 0100011

Sol. 8. **(a)**
1st complement of 1011001100 is 0100110011
2nd complement = 1st complement + 1
= 0100110011 + 0100110100

Sol. 9. **(c)**
$(101.101)_2 = (?)_{10}$
$=1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 + 1 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3}]_{10}$
$=[4+1+0.625]_{10} = (5..625)_{10}$

Sol. 10. **(b)**
1 1 0 1 1
+ 1 0 1 1 0 1 1 0
+ 1 0 0 1 1 x 0 y
----
1 0 1 1 0 1 1 0 1
----
y = 0 ,x = 1

Sol. 11. **(b)**
1 0 1 0
× 1 1 1
----
1 0 1 0
1 0 1 0 ×
1 0 1 0 × ×
----
1 0 0 0 1 1 0

$(1000110)_2 = (70)_2$

Sol. 12. **(c)**
1 1 1 1
+ 1 0 0 1
----
11 0 0 0
- 1 0 1 0

= 1 1 1 0

Sol. 13. **(c)**
1 1 0 1 0 0 1 0 1 1
- 1 1 0 0 1 1 0 0 1 1

= 1 1 0 0 0

Sol. 14. **(c)**

2	999	1
2	499	1
2	249	1
2	124	0
2	62	0
2	31	1
2	15	1
2	7	1
2	3	1
	1	1

$(999)_2=(1111100111)_2$

Sol. 15. **(c)**
1111
+ 1001

= 11000
−1010

= 1110

Sol. 16. **(d)**
For integer part of 57.375 i.e., $(57)_{10}$

2	32	0
2	16	0
2	8	0
2	4	0
2	2	0
	1	1

$\therefore (32)_{10} = (100000)_2$
For after decimal part of 32.25 i.e., $(0.25)_{10}$

Now,	Binary
0.25 × 2 = 0.50	0
0.50 × 2 = 1.00	1

$(0.25)_{10} = (10)_2$
$\therefore (32.25)_{10} = (100000.01)_2$

Sol. 17. **(c)**
For integer part of 13.625 i.e., $(13)_{10}$

2	13	1
2	6	0
2	3	1
2	1	1

$\therefore (13)_{10} = (1101)_2$
For after decimal part of 32.625 i.e., $(0.625)_{10}$
Now,

	Binary
0.625 × 2 = 1.25	1
0.25 × 2 = 0.50	0
0.50 × 2 = 1.00	1

$(0.625)_{10} = (0.101)_2$
$\therefore (13.625)_{10} = (1101.101)_2$

Sol. 18. **(c)**
$(0.0011)_2 = 2^{-1} \times 0 + 2^{-2} \times 0 + 2^{-3} \times 1 + 2^{-4} \times 1 = \frac{1}{8} + \frac{1}{16} = \frac{3}{16}$

Sol. 19. **(d)**
$\therefore$
$$\frac{(0.101)_2^{(10)_2} + (0.011)_2^{(11)_2}}{(0.101)_2^{(10)_2} - (0.101)_2^{(01)_2}(0.011)_2^{(01)_2} + (0.011)_2^{(10)_2}}$$
$$= \frac{\left(\frac{5}{8}\right)^3 + \left(\frac{3}{8}\right)^3}{\left(\frac{5}{8}\right)^2 - \left(\frac{5}{8}\right)\left(\frac{3}{8}\right) + \left(\frac{3}{8}\right)^2}$$
$$= \frac{5}{8} + \frac{3}{8} = \frac{8}{8} = 1 = (1)_2$$

Sol. 20. **(b)**
Given, $x = (1101)_2 = 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$
=8 + 4 + 1 = 13
and $y = (110)_2 = 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$
= 4 + 2 = 6
$\therefore x^2 - y^2 = (13)^2 - (6)^2 = 169 - 36 = 133$
Now,

2	133	
2	66	1
2	33	0
2	16	1
2	8	0
2	4	0
2	2	0
	1	0

$\therefore 133 = (10000101)_2$

NDA PYQ

1. The number 10101111 in binary system is represented in decimal system by which one of the following numbers?
(a) 157 (b) 175
(c) 571 (d) 751
[NDA 2011 (1)]

2. In the binary system of numbers let a = 00111 and b = 01110, then in decimal system what is b/a equal to?
(a) 1 (b) 2
(c) 4 (d) 5
[NDA 2011(1)]

3. In binary system decimal number 0.3 can be expressed as:
(a) $(0.01001....)_2$ (b) $(0.10110....)_2$
(c) $(0.11001....)_2$ (d) $(0.10101....)_2$
[NDA 2011(2)]

4. The number 292 in decimal system is expressed in binary system by:
(a) 10001010 (b) 100010001
(c) 100100100 (d) 10101000
[NDA 2012(1)]

5. What is the decimal number representation of the binary number $(11101.001)_2$?
(a) 30.125 (b) 29.025
(c) 29.125 (d) 28.025
[NDA 2012(1)]

6. Convert the decimal number $(57.375)_{10}$, into binary number
(a) $(111001.011)_2$ (b) $(100111.110)_2$
(c) $(110011.101)_2$ (d) $(111011.011)_2$
[NDA 2012(2)]

7. The decimal representation of the number $(1011)_2$ in binary system is:
(a) 5 (b) 7
(c) 9 (d) 11
[NDA 2012(2)]

8. The binary representation of the decimal number 45 is:
(a) 110011 (b) 101010
(c) 110110 (d) 101101
[NDA 2013(1)]

9. A number in binary system is 110001. It is equal to which one of the following numbers in decimal system?
(a) 45 (b) 46
(c) 48 (d) 49
[NDA 2013(2)]

10. The number 83 is written in the binary system as:
(a) 100110 (b) 101101
(c) 1010011 (d) 110110
[NDA 2013(2)]

11. The number 251 in decimal system is expressed in binary system by:
(a) 11110111 (b) 11111011
(c) 11111101 (d) 11111110
[NDA 2014(1)]

12. What is the sum of two numbers $(11110)_2$ and $(1010)_2$?
(a) $(101000)_2$ (b) $(110000)_2$
(c) $(100100)_2$ (d) $(101100)_2$
[NDA 2014(1)]

13. What is $(1001)_2$ equal to?
(a) $(5)_{10}$ (b) $(9)_{10}$
(c) $(17)_{10}$ (d) $(11)_{10}$
[NDA 2014(2)]

14. If $(11101011)_2$ is converted to decimal system, then the resulting number is:
(a) 235 (b) 175
(c) 160 (d) 126
[NDA 2015(1)]

15. The decimal number $(127.25)_{10}$ when converted to binary number takes the form.
(a) $(1111111.11)_2$ (b) $(1111110.01)_2$
(c) $(1110112.11)_2$ (d) $(1111111.01)_2$
[NDA 2015(1)]

16. What is $(1000000001)_2 - (0.0101)_2$ equal to?
(a) $(512.6775)_{10}$ (b) $(512.6875)_{10}$
(c) $(512.6975)_{10}$ (d) $(512.0909)_{10}$
[NDA-2015(2)]

17. What is the binary equivalent of the decimal number 0.3125?
(a) 0.0111 (b) 0.1010
(c) 0.0101 (d) 0.1101
[NDA (I) 2016]

18. If the number 235 in decimal system is converted into binary system, then what is the resulting number?
(a) $(11110011)_2$ (b) $(11101011)_2$
(c) $(11110101)_2$ (d) $(11011011)_2$
[NDA (II) 2016]

19. In the binary equation $(1p101)_2 + (10q1)_2 = (100r\ 00)_2$ where p, q and r are binary digits, what are the possible values of p,q and r respectively?
(a) 0,1,0 (b) 1,1,0
(c) 0,0,1 (d) 1,0,1
[NDA (I) 2017]

20. The remainder and the quotient of the binary division $(101110)_2 \div (110)_2$ are respectively.
(a) $(111)_2$ and $(100)_2$ (b) $(100)_2$ and $(111)_2$
(c) $(101)_2$ and $(101)_2$ (d) $(100)_2$ and $(100)_2$
[NDA (II) 2017]

21. The binary number expression of the decimal number 31 is:
(a) 1111 (b) 10111
(c) 11011 (d) 11111
[NDA (I) 2018]

22. The sum of the binary numbers (11011) $(10110110)_2$ and $(10011x0y)_2$ is the binary number $(101101101)_2$. What are the value of x and y ?
(a) x = 1, y = 1 (b) x = 1, y = 0
(c) x = 0, y = 1 (d) x = 0, y = 0
[NDA (II) 2018]

23. The binary number is represented by $(cdccddcccddd)_2$, where c > d what is its decimal equivalent ?
(a) 1848 (b) 2048
(c) 2842 (d) 2872
[NDA (II) 2019]

24. The number $(1101101 + 1011011)_2$ can be written in decimal system as
(a) $(198)_{10}$ (b) $(199)_{10}$
(c) $(200)_{10}$ (d) $(201)_{10}$
[NDA 2020]

25. What is $(1110011)_2 \div (10111)_2$ equal to?
(a) $(101)_2$ (b) $(1001)_2$
(c) $(111)_2$ (d) $(1011)_2$
[NDA 2022 (I)]

26. If $x^3 + y^3 = (100010111)_2$ and $x + y = (11111)_2$ then what is $(x - y)^2 + xy$ equal to?
(a) $(1101)_2$ (b) $(1001)_2$
(c) $(1011)_2$ (d) $(1111)_2$
[NDA 2022 (I)]

27. What is the binary number equivalent to decimal number 1011?
(a) 1011 (b) 111011
(c) 11111001 (d) 1111110011
[NDA 2023 (1)]

28. If $x = (1111)_2$, $y = (1001)_2$ and $z = (110)_2$, then what is $x^3 - y^3 - z^3 - 3xyz$ equal to?
(a) $(1111001)_2$ (b) $(1001111)_2$
(c) $(1)_2$ (d) $(0)_2$
[NDA 2025 (1)]

29. What is the sum of the binary numbers $(101101101)_2$ and $(100011)_2$
(a) $(110010000)_2$ (b) $(110001000)_2$
(c) $(110000100)_2$ (d) $(100100000)_2$
[NDA 2025 (2)]

ANSWER KEY

1.	b	**2.**	b	**3.**	a	**4.**	c	**5.**	c	**6.**	a	**7.**	d	**8.**	d	**9.**	d	**10.**	c
11.	b	**12.**	a	**13.**	b	**14.**	a	**15.**	d	**16.**	b	**17.**	c	**18.**	b	**19.**	a	**20.**	b
21.	d	**22**	b	**23.**	d	**24.**	c	**25.**	a	**26.**	b	**27.**	d	**28.**	d	**29.**			

Solutions

Sol. 1. (b)
The binary number is

1	0	1	0	1	1	1	1
↓	↓	↓	↓	↓	↓	↓	↓
2^7		2^5		2^3	2^2	2^1	2^0

The decimal number is = 128 + 32 + 8 + 4 + 2 + 1 = 175

Sol. 2. (b)
$a = (00111)_2 = (7)_{10}$
$b = (01110)_2 = (14)_{10}$
b/a = 14/7 = 2

Sol. 3. (a)
Given, decimal number = $(0.3)_{10}$
$\Rightarrow 0.3 \times 2 = 0.6$ 0
$\Rightarrow 0.6 \times 2 = 1.2$ 1
$\Rightarrow 0.2 \times 2 = 0.4$ 0
$\Rightarrow 0.4 \times 2 = 0.8$ 0
$\Rightarrow 0.8 \times 2 = 1.6$ 1
So, its binary equivalent is $(0.01001.....)_2$

Sol. 4. (c)

2	292	0
2	146	0
2	73	1
2	36	0
2	18	0
2	9	1
2	4	0
2	2	0
2	1	1

$(292)_2 = (100100100)_2$

Sol. 5. (c)
$1 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 + 0 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3}$
$= 16 + 8 + 4 + 0 + 1 + 0 + 0 + (1/8)$
$= 29 + (233/8) = (29.125)_{10}$

Sol. 6. (a)
For integer part of 57.375 i.e., $(57)_{10}$

2	57	
2	28	1
2	14	0
2	7	0
2	3	1
	1	1

$\therefore (57)_{10} = (111011)_2$
For after decimal part of 57.375 i.e., $(0.375)_{10}$
Now, Binary
$0.375 \times 2 = 0.75$ 0
$0.75 \times 2 = 1.5$ 1
$0.5 \times 2 = 1.0$ 1
$(0.375)_{10} = (0.011)_2$
$\therefore (57.375)_{10} = (111011.011)_2$

Sol. 7. (d)
$(1011)_2 = 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$
$= 8 + 0 + 2 + 1 = 11$

Sol. 8. (d)

2	45	1
2	22	0
2	11	1
2	5	1
2	2	0
	1	1

$\therefore$ Binary representation of $(45)_{10} = (101101)_2$

Sol. 9. (d)
$(110001)_2 = 1 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$
$= 32 + 16 + 0 + 0 + 0 + 1 = (49)_{10}$
So, the required decimal number is 49

Sol. 10. (c)

2	83	1
2	41	1
2	20	0
2	10	0
2	5	1
2	2	0
	1	1

$\therefore (83)_{10} = (1010011)_2$
Which is required binary form.

Sol. 11. (b)

2	251	1
2	125	1
2	62	0
2	31	1
2	15	1
2	7	1
2	3	1
	1	

$\therefore (251)_{10} = (11111011)_2$

Sol. 12. (a)
```
  11110
+ 1010
--------------
 101000
```

Sol. 13. (b)
$(1001)_2 = (2^3 \times 1 + 2^2 \times 0 + 2^1 \times 0 + 2^0 \times 1)_{10}$
$= (8+1)_{10} = (9)_{10}$

Sol. 14. (a)
$(11101011)_2 = (?)_{10}$
$= [7 \times 2^7 + 1 \times 2^6 + 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0]_{10}$
$= [128+64+32+0+8+0+2+1]_{10} = (235)_{10}$

Sol. 15. (d)
$(127.25)_{10} = (?)_2$

2	127	1
2	63	1
2	31	1
2	15	1
2	7	1
2	3	1
2	1	1
	0	1

Here Binary
$0.25 \times 2 = 0.50$ 0
$0.50 \times 2 = 1.00$ 1
$\therefore (127.25)_{10} = (1111111.01)_2$

Sol. 16. (b)
$(1000000001)_2 - (0.0101)_2$

$=(1 \times 2^9 + 0 \times 2^8 + ...+ 0 \times 2^1 + 1 \times 2^0)$
$-(0 \times 2^{-1} + 1 \times 2^{-2} + 0 \times 2^{-3} + 1 \times 2^{-4})$
$=512 + 1 - (1/4 + 1/16)$
$=513 - \frac{5}{16} = 513-0.3125 = (512.6875)_{10}$

Sol. 17. **(c)**
$0.3125 \times 2 = 0.6250$ 1
$0.6250 \times 2 = 1.3500$ 1
$0.3500 \times 2 = 0.7000$ 0
$0.7000 \times 2 = 1.4000$ 1
$(0.3125)_{10} = (0.0101)_2$

Sol. 18. **(b)**

2	2 3 5	
2	1 2 7	1
2	5 8	1
2	2 9	0
2	1 4	1
2	7	0
2	3	1
2	1	1
	0	1

$\therefore (235)_{10} = (11101011)_2$

Sol. 19. **(a)**
```
   1 p 1 0 1
+    1 0 q 1
---------------------
=  1 0 0 r 0 0
```
p = 0, q = 1, r = 0

Sol. 20. **(b)**
$(101110)_2 \equiv (46)_{10}$ and $(110)_2 \equiv (6)_{10}$ when $46 \div 5$, quotient = 7 and remainder = 4
$\therefore$ Remainder $= (4)_{10} \equiv (100)_2$ and quotient $= (7)_{10} \equiv (111)_2$

Sol. 21. **(d)**

2	31	1
2	15	1
2	7	1
2	3	1
2	1	1
	0	

$\therefore$ Binary expression of decimal number 31 = 11111

Sol. 22. **(b)**
```
  101101101
– 10110110
-------------------
   10110111
–     11011
--------------------
   10011100
```
$\Rightarrow x = 1, y = 0$

Sol. 23. **(d)**
Given that c > d i.e. c = 1 and d = 0
$(cdccddcccddd)_2 = (101100111000)_2$
$=(1 \times 2^{11} + 1 \times 2^9 + 1 \times 2^8 + 1 \times 2^5 + 1 \times 2^4 + 1 \times 2^3) = 2872$

Sol. 24. **(c)**
```
   1101101
 + 1011011
-------------
  11001000   = (200)₂
```

Sol. 25. **(a)**
```
10111 ) 1110011 ( 101
        10111
        -----
         10111
             0
         -----
         10111
         10111
         -----
             X
```
Answer $= (101)_2$

Sol. 26. **(b)**
$(x-y)^2 + xy$
$= x^2 + y^2 - 2xy + xy$
$= x^2 + y^2 - xy$
$= \frac{x^3 + y^3}{x + y}$
```
11111 ) 100010111 ( 1001
        11111
        -----
          111
            0
        -----
          1111
             0
        -----
          1111
          1111
        -----
             X
```
Answer $= (1001)_2$

Sol. 27. **(d)**

2	1011	1
2	505	1
2	252	0
2	126	0
2	63	1
2	31	1
2	15	1
2	7	1
2	3	1
2	1	1
	0	

$(1011)_{10} = (1111110011)_2$

Sol. 28. **(d)**
$x^3 - y^3 - z^3 - 3xyz$
$= (x - y - z)(x^2 + y^2 + z^2 - xy - yz + zx)$
Here $x = (1111)_2 = (15)_{10}$,
$y = (1001)_2 = (9)_{10}$,
$z = (110)_2 = (6)_{10}$
$(x - y - z) = 0$
So $x^3 - y^3 - z^3 - 3xyz = 0$

Sol. 29. **(a)**
```
  1 0 1 1 0 1 1 0 1
+       1 0 0 0 1 1
--------------------
= 1 1 0 0 1 0 0 0 0
```

You can access the solutions of PYQ through 3 Youtube video:
1. Binary numbers 2011 to 2015 | NDA mathematics previous year questions by Sandeep Brar
2. Binary numbers 2016 to 2020 | NDA mathematics previous year questions by Sandeep Brar
3. Binary numbers 2021 to 2025 | NDA mathematics previous year questions by Sandeep Brar